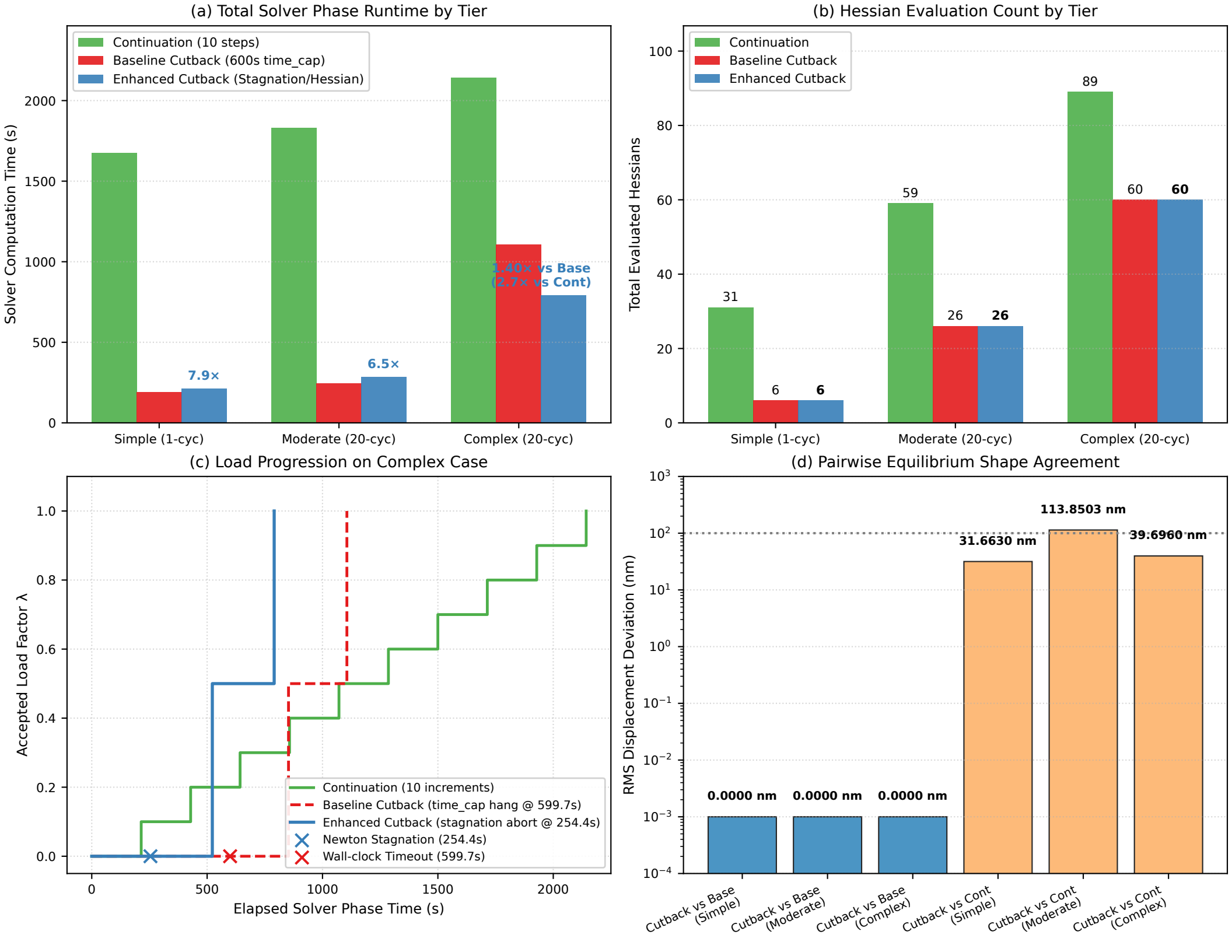
(a) Total Solver Phase Runtime by Tier
Continuation (10 steps)
Baseline Cutback (600s time_cap)
Enhanced Cutback (Stagnation/Hessian)
Solver Computation Time (s)
0
500
1000
1500
2000
7.9×
6.5×
1.40× vs Base
(2.7× vs Cont)
Simple (1-cyc)
Moderate (20-cyc)
Complex (20-cyc)
(b) Hessian Evaluation Count by Tier
Continuation
Baseline Cutback
Enhanced Cutback
Total Evaluated Hessians
0
20
40
60
80
100
31
6
6
59
26
26
89
60
60
Simple (1-cyc)
Moderate (20-cyc)
Complex (20-cyc)
(c) Load Progression on Complex Case
Accepted Load Factor λ
0.0
0.2
0.4
0.6
0.8
1.0
0
500
1000
1500
2000
Elapsed Solver Phase Time (s)
Continuation (10 increments)
Baseline Cutback (time_cap hang @ 599.7s)
Enhanced Cutback (stagnation abort @ 254.4s)
Newton Stagnation (254.4s)
Wall-clock Timeout (599.7s)
(d) Pairwise Equilibrium Shape Agreement
RMS Displacement Deviation (nm)
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-2}
10^{-3}
10^{-4}
0.0000 nm
0.0000 nm
0.0000 nm
31.6630 nm
113.8503 nm
39.6960 nm
Cutback vs Base (Simple)
Cutback vs Base (Moderate)
Cutback vs Base (Complex)
Cutback vs Cont (Simple)
Cutback vs Cont (Moderate)
Cutback vs Cont (Complex)